

Introduction to Abstract Algebra II: Homework 7

Due on March 4, 2026 at 23:59

Victor Ostrik 10:00

Hashem A. Damrah

UO ID: 952102243

Exercise 34.10. Factor the polynomial $4x^2 - 4x + 8$ into a product of irreducibles viewing it as an element of the integral domain $\mathbb{Z}[x]$; of the integral domain $\mathbb{Q}[x]$; of the integral domain $\mathbb{Z}_{11}[x]$.

Solution. We factor the polynomial $f(x) = 4x^2 - 4x + 8$ in three different polynomial rings.

- $\mathbb{Z}[x]$: First factor out the greatest common divisor of the coefficients:

$$f(x) = 4(x^2 - x + 2).$$

Now check whether the quadratic factors further over $\mathbb{Z}$. Its discriminant is

$$\Delta = (-1)^2 - 4(1)(2) = 1 - 8 = -7.$$

Since the discriminant is negative, the quadratic has no rational (hence no integer) roots, so it is irreducible over $\mathbb{Z}$.

- $\mathbb{Q}[x]$: The discriminant is still -7 , which is not a square in $\mathbb{Q}$, so the quadratic does not factor over $\mathbb{Q}$. Also, nonzero constants are units in $\mathbb{Q}[x]$, so the factor 4 is a unit.
- $\mathbb{Z}_{11}[x]$: Reduce the coefficients modulo 11:

$$f(x) \equiv 4x^2 + 7x + 8 \pmod{11}.$$

Compute the discriminant in $\mathbb{Z}_{11}$:

$$\Delta = 7^2 - 4(4)(8) = 49 - 128 = -79 \equiv 9 \pmod{11}.$$

Since 9 is a square in $\mathbb{Z}_{11}$ ($9 = 3^2 = 8^2$), the polynomial factors. The roots are

$$x = \frac{-7 \pm 3}{2 \cdot 4}.$$

Therefore, the roots are $x = 5$ and $x = 7$. Hence, we can factor the polynomial to $4x^2 - 4x + 8 = 4(x - 5)(x - 7)$

Thus, we have

$$\begin{aligned} \mathbb{Z}[x] : f(x) &= 4(x^2 - x + 2) \\ \mathbb{Q}[x] : f(x) &= 4x^2 - 4x + 8 \\ \mathbb{Z}_{11}[x] : f(x) &= 4(x - 5)(x - 7). \end{aligned} \quad \square$$

Exercise 34.25. Prove that if p is a prime in an integral domain D , then p is an irreducible.

Solution. Let D be an integral domain and suppose $p \in D$ is a prime element. By definition, p is prime if $p \neq 0$, p is not a unit, and whenever p divides ab , then p divides either a or b . We must show that p is irreducible. Suppose $p = ab$ for some $a, b \in D$. Since $p = ab$, we have $p \mid ab$. Because p is prime, it follows that $p \mid a$ or $p \mid b$.

If $p \mid a$, then there exists $c \in D$ such that $a = pc$. Substituting into $p = ab$ gives $p = (pc)b = p(cb)$. Since D is an integral domain, we may cancel $p \neq 0$, yielding $1 = cb$. Thus b is a unit in D . If $p \mid b$, then $b = pc$ for some $c \in D$. Substituting into $p = ab$ gives $p = a(pc) = p(ac)$. Canceling p (since D is an integral domain) gives $1 = ac$. In either case, one of the factors a or b is a unit. Therefore p is irreducible.

Thus, every prime element in an integral domain is irreducible. $\square$

Exercise 34.26. Prove that if p is an irreducible in a UFD, then p is a prime.

Solution. Let D be a unique factorization domain (UFD) and let $p \in D$ be irreducible. We must prove that p is prime. Suppose $p \mid ab$. Then there exists some $c \in D$ such that $ab = pc$. Since D is a UFD, every element has a factorization into irreducibles that is unique up to order and multiplication by units.

Write the factorizations of a and b into irreducibles:

$$a = up_1p_2 \cdots p_m, \quad b = vq_1q_2 \cdots q_n,$$

where u, v are units in D , and each p_i, q_j is irreducible. Then

$$ab = (uv)p_1p_2 \cdots p_mq_1q_2 \cdots q_n.$$

On the other hand, pc is a factorization of ab in which p appears as one of the irreducible factors. Since factorizations in a UFD are unique up to associates and ordering, the irreducible element p must be associate to one of the irreducible factors appearing in the factorization of ab . Thus p is associate to some p_i or to some q_j .

If p is associate to p_i , then $p \mid a$. If p is associate to q_j , then $p \mid b$. Therefore, whenever $p \mid ab$, we have $p \mid a$ or $p \mid b$. Hence p is prime.

Thus every irreducible element in a unique factorization domain is prime. $\square$

Exercise 34.27. For a commutative ring R with unity show that the relation $a \sim b$ if a is an associate of b (that is, if $a = bu$ for u a unit in R) is an equivalence relation on R .

Solution. Let R be a commutative ring with unity. Define a relation on R by

$$a \sim b \quad \text{if and only if} \quad a = bu$$

for some unit $u \in R$. We show that this relation is an equivalence relation by verifying that it is reflexive, symmetric, and transitive.

For any $a \in R$, we have $a = a \cdot 1$. Since 1 is a unit in R , it follows that a is an associate of itself. Hence $a \sim a$. Suppose $a \sim b$. Then $a = bu$ for some unit $u \in R$. Since u is a unit, it has an inverse u^{-1} in R . Multiply both sides by u^{-1} to get $b = au^{-1}$. Because u^{-1} is also a unit, this shows that b is an associate of a , so $b \sim a$. Suppose $a \sim b$ and $b \sim c$. Then there exist units $u, v \in R$ such that $a = bu$, $b = cv$. Substituting gives $a = (cv)u = c(vu)$. Since the product of units is again a unit, vu is a unit in R . Hence a is an associate of c , so $a \sim c$.

Thus, it is an equivalence relation on R . $\square$

Exercise 34.29. Let D be a UFD. Show that a nonconstant divisor of a primitive polynomial in $D[x]$ is again a primitive polynomial.

Solution. Let D be a UFD and let $f(x) \in D[x]$ be a primitive polynomial. Recall that a polynomial is *primitive* if the greatest common divisor of its coefficients is 1.

Suppose $g(x) \in D[x]$ is a nonconstant polynomial such that $g(x) \mid f(x)$. Then there exists $h(x) \in D[x]$ with $f(x) = g(x)h(x)$. Assume for contradiction that $g(x)$ is not primitive. Then the coefficients of $g(x)$ have a nonunit common divisor. Since D is a UFD, there exists an irreducible element $p \in D$ such that p divides every coefficient of $g(x)$. Thus we may write $g(x) = pg_1(x)$, for some $g_1(x) \in D[x]$. Substituting into the factorization of $f(x)$ gives $f(x) = pg_1(x)h(x)$. This implies that p divides every coefficient of $f(x)$, since multiplying by p multiplies each coefficient by p . Hence p is a common divisor of all coefficients of $f(x)$. But this contradicts the assumption that $f(x)$ is primitive, since a primitive polynomial has no nonunit common divisor of its coefficients. Therefore our assumption was false, and $g(x)$ must be primitive.

Hence every nonconstant divisor of a primitive polynomial in $D[x]$ is also primitive. $\square$

Exercise 34.30. Show that in a PID, every proper ideal is contained in a maximal ideal. [Hint: Use Lemma 34.10.]

Solution. Let R be a PID and let I be a proper ideal of R . Since R is a PID, I is principal, so there exists $a \in R$ such that $I = \langle a \rangle$. Because I is proper, a is not a unit. Factor a into irreducibles. Since every PID is a UFD, we may write $a = p_1 p_2 \cdots p_n$, where each p_i is irreducible. Choose one of these irreducible factors, say p_1 . Then $p_1 \mid a$, so $a = p_1 b$, for some $b \in R$. This implies $\langle a \rangle \subseteq \langle p_1 \rangle$, since every multiple of a is also a multiple of p_1 . In a PID, every irreducible element generates a maximal ideal. Hence $\langle p_1 \rangle$ is a maximal ideal of R . Thus we have

$$I = \langle a \rangle \subseteq \langle p_1 \rangle,$$

where $\langle p_1 \rangle$ is a maximal ideal. Therefore I is contained in a maximal ideal. Thus, every proper ideal in a PID is contained in a maximal ideal. $\square$

Exercise 35.2. The function ν for $\mathbb{Z}[x]$ given by $\nu(f(x)) = (\text{degree of } f(x))$ for $f(x) \in \mathbb{Z}[x]$, $f(x) \neq 0$.

Solution. Consider the function $\nu(f(x)) = \deg(f(x))$ for nonzero $f(x) \in \mathbb{Z}[x]$. We show that this function does *not* satisfy the Euclidean property. Take

$$a(x) = x, \quad b(x) = 2.$$

Suppose there exist $q(x), r(x) \in \mathbb{Z}[x]$ such that

$$x = 2q(x) + r(x),$$

with either $r(x) = 0$ or $\deg(r) < \deg(2)$. Since $\deg(2) = 0$, the condition $\deg(r) < 0$ implies that $r(x) = 0$. Thus the equation would require $x = 2q(x)$. However, this would imply

$$q(x) = \frac{x}{2},$$

which is not an element of $\mathbb{Z}[x]$ because its coefficients are not integers. Hence no such $q(x)$ exists. Therefore it is impossible to write $x = 2q(x) + r(x)$ with $q(x), r(x) \in \mathbb{Z}[x]$ and $\deg(r) < \deg(2)$. Thus the function $\nu(f) = \deg(f)$ does not satisfy the Euclidean division property in $\mathbb{Z}[x]$. Consequently, this function is not a Euclidean valuation for $\mathbb{Z}[x]$, and $\mathbb{Z}[x]$ is not a Euclidean domain with respect to degree. $\square$

Exercise 35.8. Following the idea of Exercise 6 and referring to Exercise 7, express the positive gcd of 49,349 and 15,555 in $\mathbb{Z}$ in the form $\lambda(49,349) + \mu(15,555)$ for $\lambda, \mu \in \mathbb{Z}$.

Solution. We use the Euclidean algorithm to compute $\gcd(49,349, 15,555)$ and then express it as a linear combination.

$$\begin{aligned} - 49349 &= 3(15555) + 2684, \\ - 15555 &= 5(2684) + 2135, \\ - 2684 &= 1(2135) + 549, \\ - 2135 &= 3(549) + 488, \\ - 549 &= 1(488) + 61, \\ - 488 &= 8(61) + 0. \end{aligned}$$

Thus,

$$\gcd(49,349, 15,555) = 61.$$

Now, we back-substitute it. From $61 = 549 - 488$, substitute $488 = 2135 - 3(549)$ to get

$$61 = 549 - (2135 - 3(549)) = 4(549) - 2135.$$

Substitute $549 = 2684 - 2135$:

$$61 = 4(2684) - 5(2135).$$

Substitute $2135 = 15555 - 5(2684)$:

$$61 = 29(2684) - 5(15555).$$

Substitute $2684 = 49349 - 3(15555)$:

$$61 = 29(49349) - 92(15555).$$

Thus $\lambda = 29$ and $\mu = -92$. □

Exercise 35.9. Find the gcd of

$$x^{10} - 3x^9 + 3x^8 - 11x^7 + 11x^6 - 11x^5 + 19x^4 - 13x^3 + 8x^2 - 9x + 3,$$

and

$$x^6 - 3x^5 + 3x^4 - 9x^3 + 5x^2 - 5x + 2,$$

in $\mathbb{Q}[x]$.

Solution. Since $\mathbb{Q}[x]$ is a Euclidean domain with Euclidean function $\nu(f) = \deg(f)$, we can compute the gcd using the Euclidean algorithm for polynomials. Repeatedly dividing the higher-degree polynomial by the lower-degree one and replacing the pair by the divisor and remainder eventually yields a remainder of 0. The last nonzero remainder is the gcd (up to multiplication by a nonzero constant). Carrying out the Euclidean algorithm for $f(x)$ and $g(x)$ in $\mathbb{Q}[x]$ yields the final nonzero remainder

$$x^3 + 2x - 1.$$

The polynomial $x^3 + 2x - 1$ is already monic, so no normalization is needed. Therefore,

$$\gcd(f(x), g(x)) = x^3 + 2x - 1. \quad \square$$

Exercise 35.15. Let D be a Euclidean domain and let ν be a Euclidean norm on D . Show that if a and b are associates in D , then $\nu(a) = \nu(b)$.

Solution. Let D be a Euclidean domain with Euclidean norm ν . Suppose $a, b \in D$ are associates. By definition of associates, there exists a unit $u \in D$ such that $a = bu$. Since u is a unit, it has a multiplicative inverse $u^{-1} \in D$. Thus we also have $b = au^{-1}$. In a Euclidean domain, the norm satisfies the property that for nonzero elements $x, y \in D$, $\nu(x) \leq \nu(xy)$. Applying this with $x = b$ and $y = u$, we obtain $\nu(b) \leq \nu(bu) = \nu(a)$. Similarly, applying the same property with $x = a$ and $y = u^{-1}$ gives $\nu(a) \leq \nu(au^{-1}) = \nu(b)$. Combining the two inequalities, we conclude that $\nu(a) = \nu(b)$. Therefore, associates in a Euclidean domain have the same Euclidean norm. □